LEC 9.2: PROPERTIES OF CT FOURIER SERIES (CTFS)

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Properties of Continuous Time Fourier Series

- **1) Linearity:**
- If $x(t) \leftrightarrow a_k$ and $y(t) \leftrightarrow b_k$
- $z(t) = Ax(t) + By(t) \leftrightarrow Z(k) = Aa_k + Bb_k$

- Comment: It is used to analyze signals which are linear combination of other signals.

- **2) Time Shift / Translation:**
- If $x(t) \leftrightarrow a_k$
- $x(t - t_o) \leftrightarrow e^{-jk\omega t_o} a_k$

- **3) Frequency Shift:**
- If $x(t) \leftrightarrow a_k$ then
- $e^{jk_o\omega_o t}x(t) \leftrightarrow a_{k-k_o}$

- **4) Time Scaling:**
- If $x(t) \leftrightarrow a_k$ then
- $x(\alpha t) \leftrightarrow a_k$

- Comment: Fourier coefficients of $x(t)$ and $x(\alpha t)$ are same but spacing between frequency components change from ω_o to $\alpha\omega_o$.

- 5) Time Differentiation:
- If $x(t) \leftrightarrow a_k$ then
- $\frac{d}{dt}x(t) \leftrightarrow jk\omega a_k$

- 6) Convolution in Time:
- If $x(t) \leftrightarrow a_k$ and $y(t) \leftrightarrow b_k$
- $x(t) * y(t) \leftrightarrow Ta_k\, b_k$

- Comment: Convolution of two periodic signals result in multiplication of their Fourier coefficients and period T

- <u>7) Multiplication:</u>
- If $x(t) \leftrightarrow a_k$ and $y(t) \leftrightarrow b_k$
- $x(t)y(t) \leftrightarrow a_k * b_k$

- <u>8) Time Integration:</u>
- If $x(t) \leftrightarrow a_k$ and
- $\int_{-\infty}^{t} x(t) \leftrightarrow \frac{1}{jk\omega} a_k$

- <u>9) Time Integration:</u>
- If $x(t) \leftrightarrow a_k$ and
- $x(-t) \leftrightarrow a_k$

Table

Table 1: **Properties of the Continuous-Time Fourier Series**

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{jk(2\pi/T)t}$$

$$a_k = \frac{1}{T}\int_T x(t)e^{-jk\omega_0 t}dt = \frac{1}{T}\int_T x(t)e^{-jk(2\pi/T)t}dt$$

Property	**Periodic Signal**	**Fourier Series Coefficients**
	$\left.\begin{matrix} x(t) \\ y(t) \end{matrix}\right\}$ Periodic with period T and fundamental frequency $\omega_0 = 2\pi/T$	a_k b_k
Linearity	$Ax(t) + By(t)$	$Aa_k + Bb_k$
Time-Shifting	$x(t - t_0)$	$a_k e^{-jk\omega_0 t_0} = a_k e^{-jk(2\pi/T)t_0}$
Frequency-Shifting	$e^{jM\omega_0 t} = e^{jM(2\pi/T)t}x(t)$	a_{k-M}
Conjugation	$x^*(t)$	a^*_{-k}
Time Reversal	$x(-t)$	a_{-k}
Time Scaling	$x(\alpha t), \alpha > 0$ (periodic with period T/α)	a_k
Periodic Convolution	$\int_T x(\tau)y(t-\tau)d\tau$	$Ta_k b_k$
Multiplication	$x(t)y(t)$	$\sum_{l=-\infty}^{+\infty} a_l b_{k-l}$

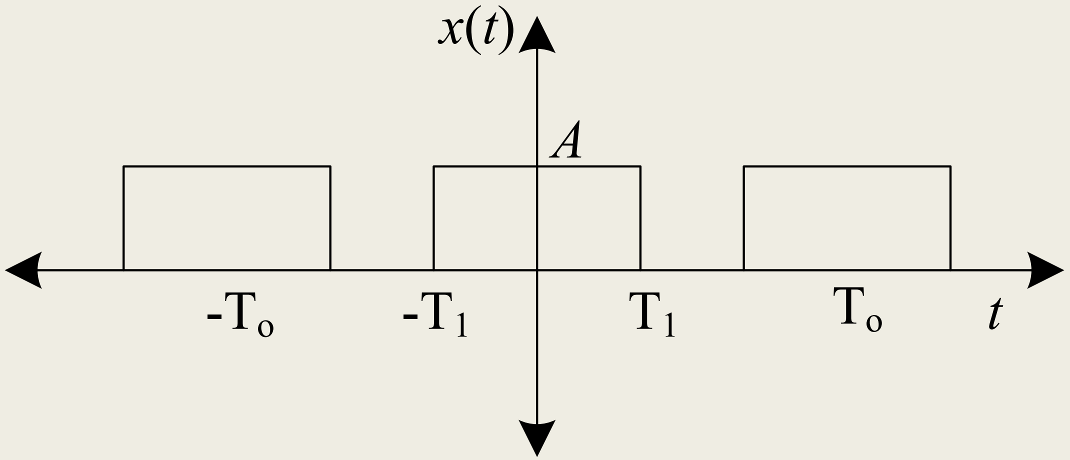
Differentiation	$\dfrac{dx(t)}{dt}$	$jk\omega_0 a_k = jk\dfrac{2\pi}{T}a_k$
Integration	$\int_{-\infty}^{t} x(t)dt$ (finite-valued and periodic only if $a_0 = 0$)	$\left(\dfrac{1}{jk\omega_0}\right) a_k = \left(\dfrac{1}{jk(2\pi/T)}\right) a_k$
Conjugate Symmetry for Real Signals	$x(t)$ real	$\begin{cases} a_k = a^*_{-k} \\ \Re e\{a_k\} = \Re e\{a_{-k}\} \\ \Im m\{a_k\} = -\Im m\{a_{-k}\} \\ \|a_k\| = \|a_{-k}\| \\ \sphericalangle a_k = -\sphericalangle a_{-k} \end{cases}$
Real and Even Signals	$x(t)$ real and even	a_k real and even
Real and Odd Signals	$x(t)$ real and odd	a_k purely imaginary and odd
Even-Odd Decomposition of Real Signals	$\begin{cases} x_e(t) = \mathcal{E}v\{x(t)\} & [x(t) \text{ real}] \\ x_o(t) = \mathcal{O}d\{x(t)\} & [x(t) \text{ real}] \end{cases}$	$\Re e\{a_k\}$ $j\Im m\{a_k\}$

Parseval's Relation for Periodic Signals

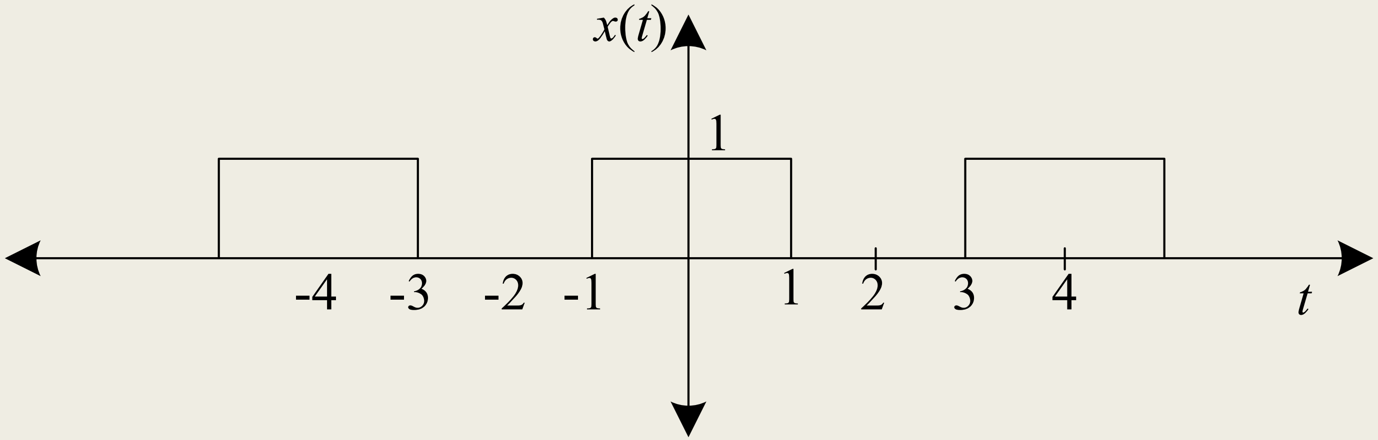
$$\frac{1}{T}\int_T |x(t)|^2 dt = \sum_{k=-\infty}^{+\infty} |a_k|^2$$

Fourier Series Coefficients of some basic signals

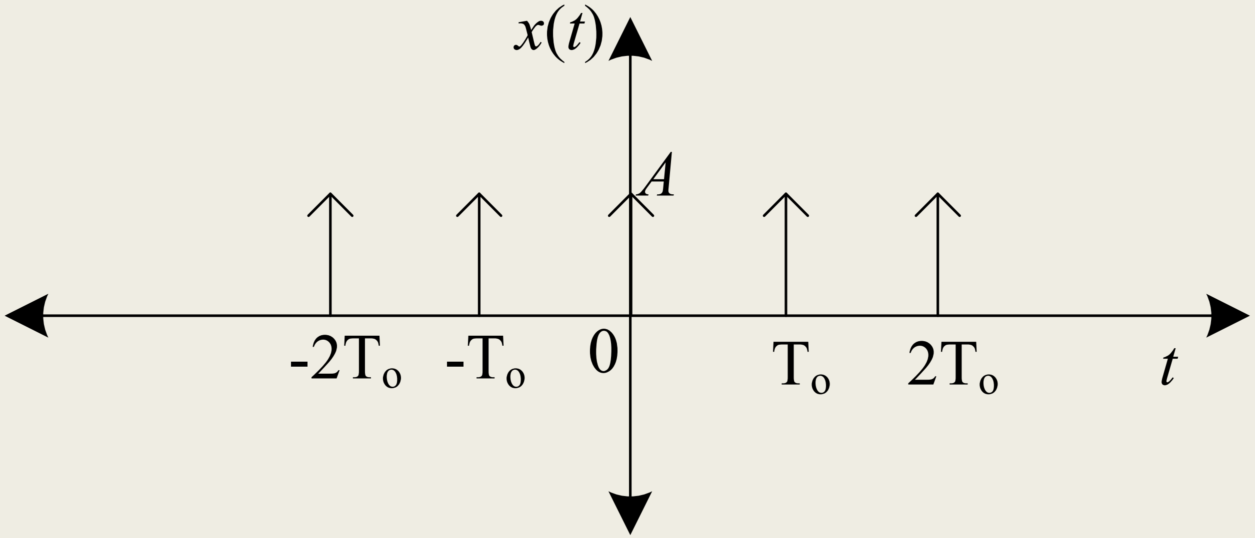
- Fourier coefficients
- $a_k = \frac{2\sin(k\omega T_1)}{k\omega T}.A$



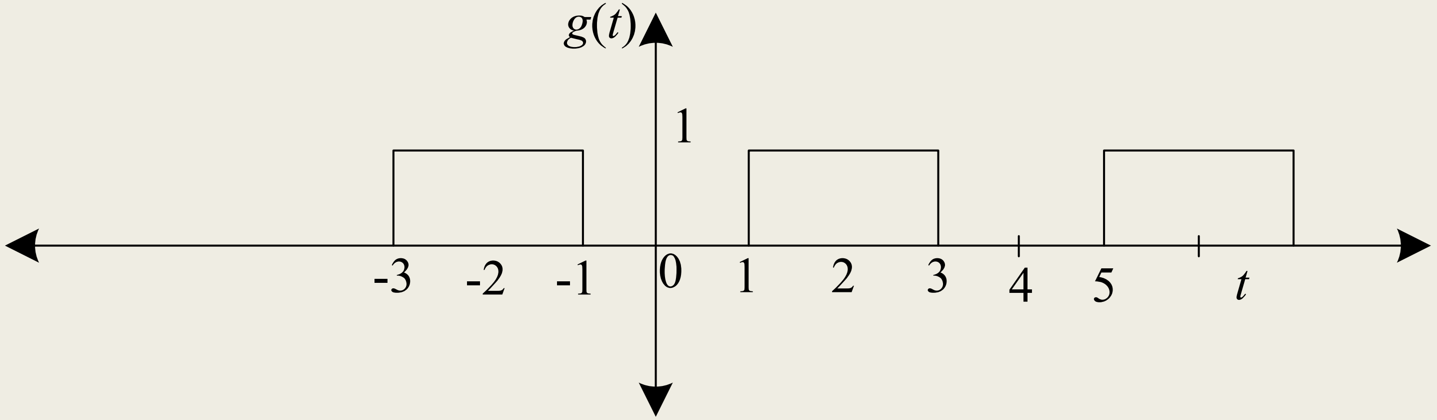
- Fourier coefficients
- $a_k = \frac{2\sin(k\omega T_1)}{k\omega T}.A$
  - $T = 4, \omega = \frac{2\pi}{T}$
  - $A = 1$
  - $T_1 = 1$

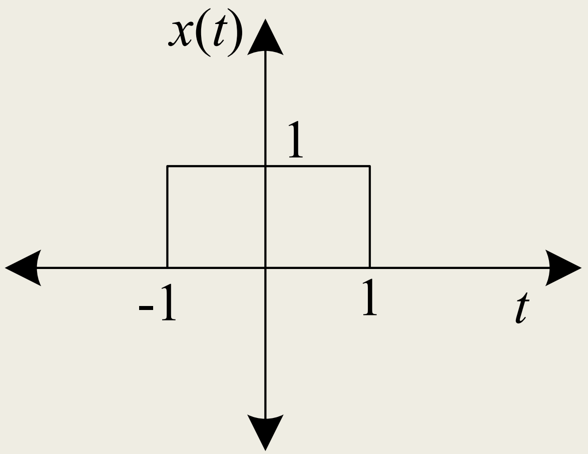


- Fourier coefficients
- $a_k = \frac{A}{T_o}$



Exp 1: Write FS expansion for the signal given below.



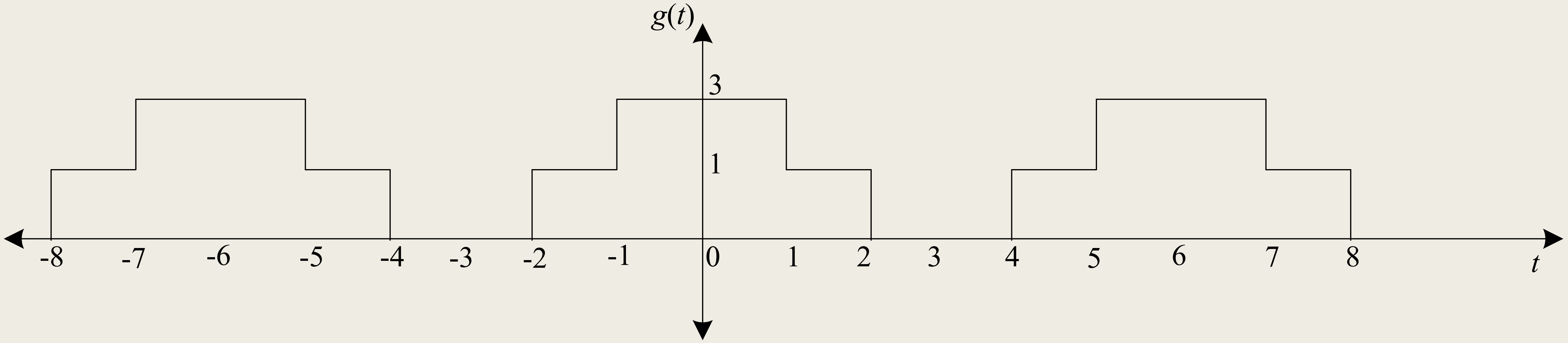


- $a_k = \frac{2\sin(k\omega T_1)}{k\omega T}.A$
- $a_k = \frac{A}{k\pi}2\sin(k\omega.1)$
- $a_k = \frac{A}{k\pi}\sin(k\omega)$
    - $\omega = \frac{2\pi}{T}$
    - $T = 5 - 1 = 4$
- $a_k = \frac{1}{k\pi}\sin\left(k\frac{\pi}{2}\right)$

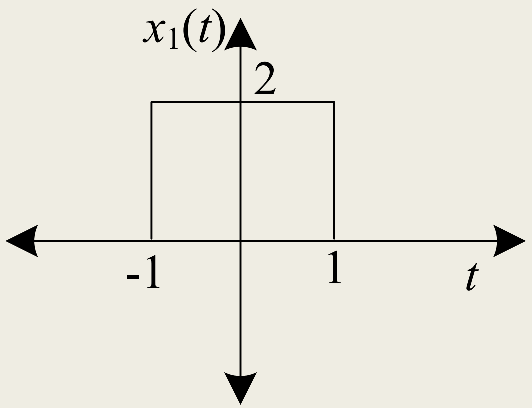
With reference to $x(t)$, gate function in $g(t)$ is shifted by '2' steps.

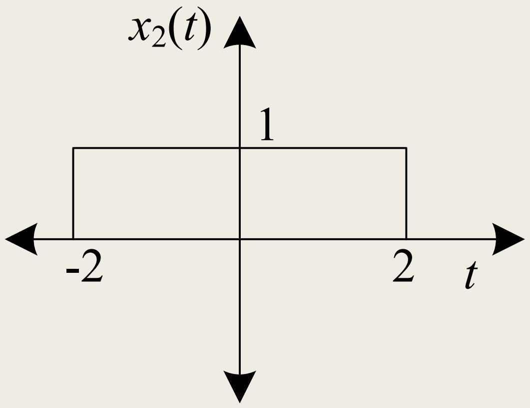
- With reference to $x(t)$, gate function in $g(t)$ is shifted by '2' steps. Thus, $t_o = 2$
- Apply shift property
- $z(t) = x(t - t_o) \quad \leftrightarrow b_k = e^{-jk\omega t_o} a_k$
- $a_k = \frac{1}{k\pi} \sin\left(k \frac{\pi}{2}\right)$
- $b_k = e^{-jk\omega t_o} a_k$
    - $t_o = 2$
    - $\omega = \frac{2\pi}{T} = \frac{2\pi}{4} = \frac{\pi}{2}$
- $b_k = e^{-jk(\frac{\pi}{2})2} a_k$
- $b_k = e^{-jk\pi} a_k$

Exp 2: Write FS expansion for the signal given below using properties

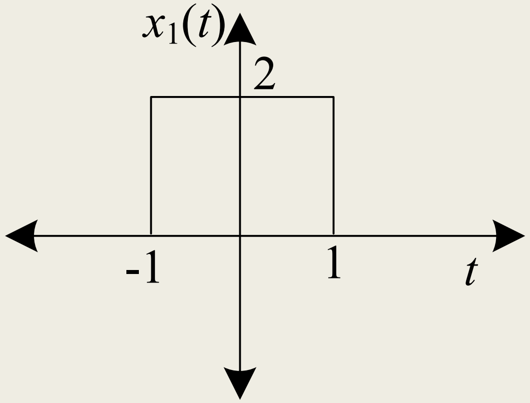


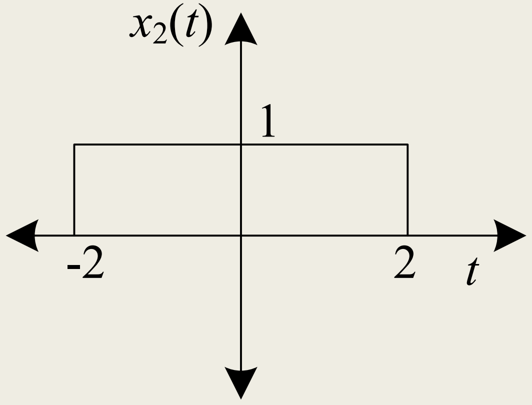
- $T_o = 6 - 0 = 6$
- $x(t)$ can be written as linear combination of two signals $x_1(t)$ and $x_2(t)$
- $x(t) = x_1(t) + x_2(t)$





- FS coefficients for gate function $x_1(t)$
- $b_k = \frac{A}{k\pi}\sin(k\omega T_1)$
    - $A = 2$
    - $\omega = \frac{2\pi}{T}$
    - $T = 6$
    - $T_1 = 1$
- $b_k = \frac{2}{k\pi}\sin\left(k\frac{2\pi}{6}\right)$
- $\quad = \frac{2}{k\pi}\sin\left(k\frac{\pi}{3}\right)$



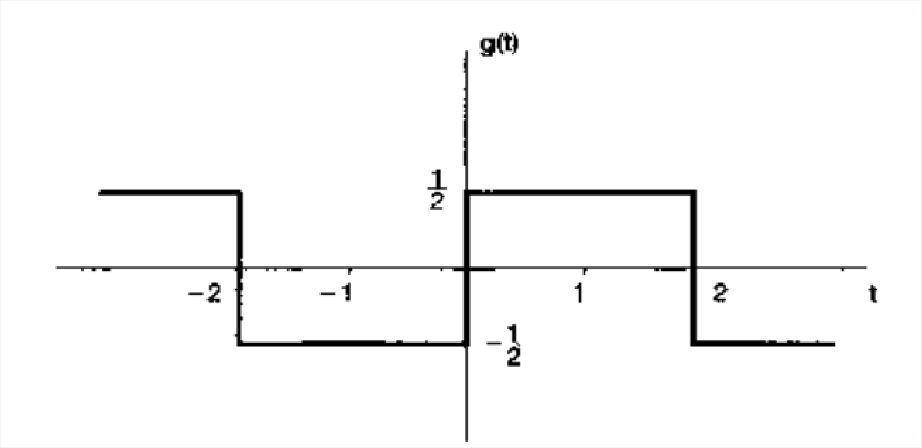


- FS coefficients for gate function $x_2(t)$
- $c_k = \frac{A}{k\pi}\sin(k\omega T_1)$
    - $A = 1$
    - $\omega = \frac{2\pi}{T_o}$
    - $T = 6$
    - $T_1 = 2$
- $c_k = \frac{1}{k\pi}\sin\left(k\frac{2\pi}{6}(2)\right)$
- $\quad = \frac{1}{k\pi}\sin\left(k\frac{2\pi}{3}\right)$

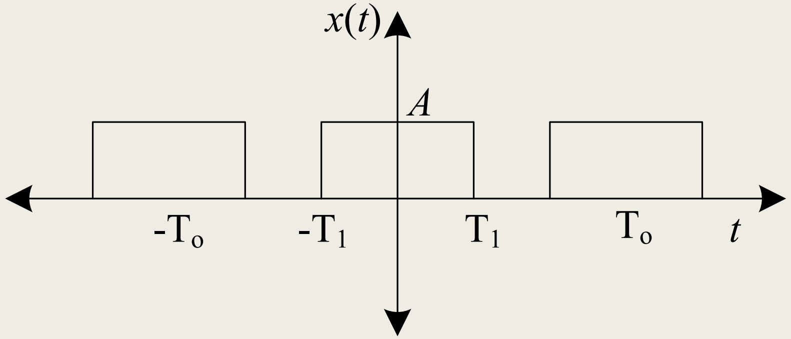
- Thus, FS coefficients for $x(t)$ i.e. a_k is simply the linear combination of b_k and c_k

- $a_k = b_k + c_k$
- $a_k = \frac{2}{k\pi} \sin\left(k\frac{\pi}{3}\right) + \frac{1}{k\pi} \sin\left(k\frac{2\pi}{3}\right)$

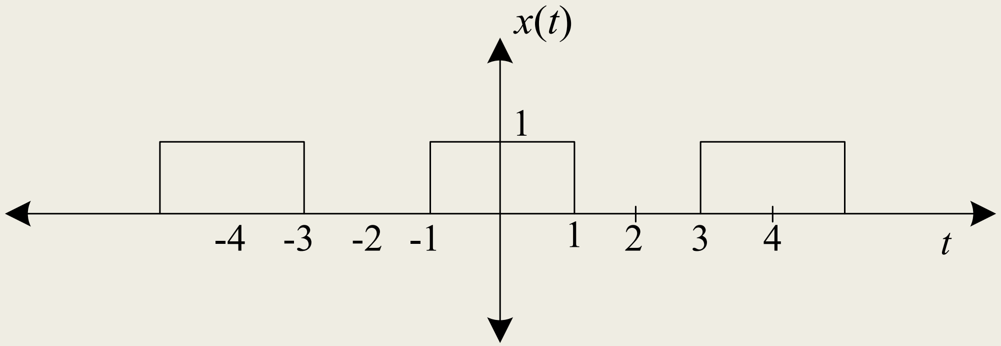
Exp 3 (3.6): Write FS expansion for the signal given below.



- We have to write this signal as a linear combination of different simple signals.
- As we know that the basic gate function has form



- Width of gate in $g(t)$ is 2, therefore put $T_1 = 1$ for the generic gate signal. Thus, Fourier coefficients for $x(t)$ are given as;
- $a_k = \frac{1}{k\pi} \sin\left(k\frac{2\pi}{4}\right)$, $a_o = \frac{A}{2}$
- $a_k = \frac{1}{k\pi} \sin\left(k\frac{\pi}{2}\right)$, , $a_o = \frac{1}{2}$



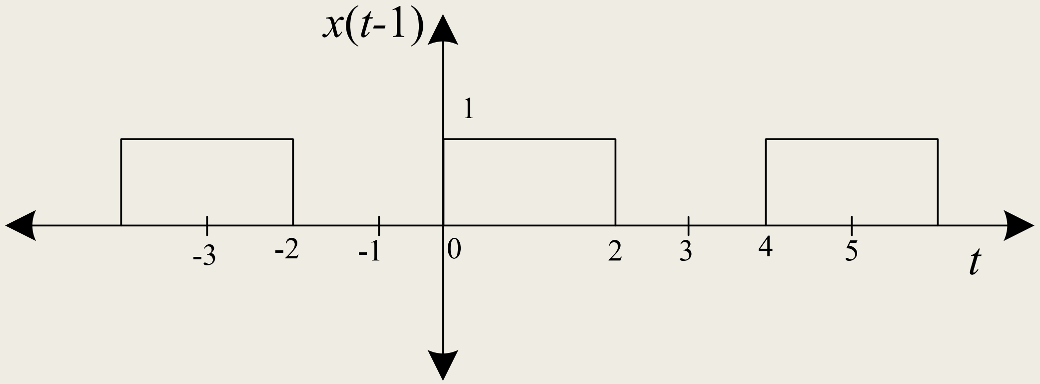
- Shift $x(t)$ by 1 step results in $x(t-1)$.

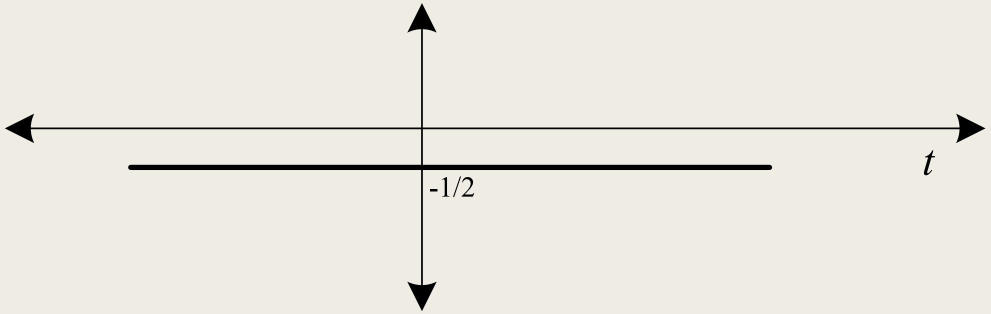
- $g(t) = x_1(t) + x_2(t)$
- $g(t) = \ x(t-1) - \frac{1}{2}$

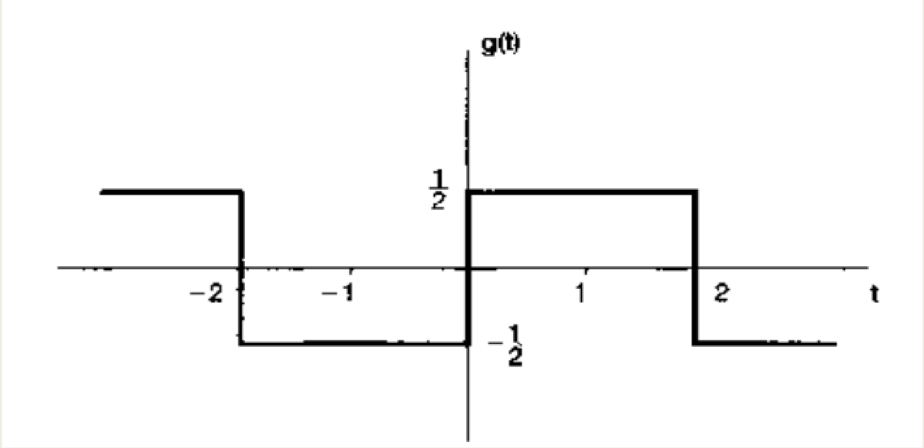
- FS coefficients for $x(t)$
- $a_k = \frac{1}{k\pi}\sin\left(k\frac{\pi}{2}\right), a_o = \frac{A}{2}$

- For $x(t-1)$, apply shift property
- $z(t) = x(t-t_o) \qquad \leftrightarrow \qquad Z(k) = e^{-jk\omega_o t_o} a_k$

- $\therefore \ x(t-1) \leftrightarrow e^{-jk\omega_o t_o} \frac{1}{k\pi}\sin\left(k\frac{\pi}{2}\right)$







- $\therefore\ x(t-1) \leftrightarrow e^{-jk\omega_o t_o}\frac{1}{k\pi}\sin\left(k\frac{\pi}{2}\right)$
- $t_o = 1$ and $\omega_o = \frac{2\pi}{4} = \frac{\pi}{2}$

- $x(t-1) \leftrightarrow e^{-jk\frac{\pi}{2}}\frac{1}{k\pi}\sin\left(k\frac{\pi}{2}\right)$

- FS coefficients for $x_1(t) = x(t-1)$ are
- $b_k = \begin{cases} e^{-jk\frac{\pi}{2}}\frac{1}{k\pi}\sin\left(k\frac{\pi}{2}\right) & k \neq 0 \\ \frac{1}{2} & k = 0 \end{cases}$

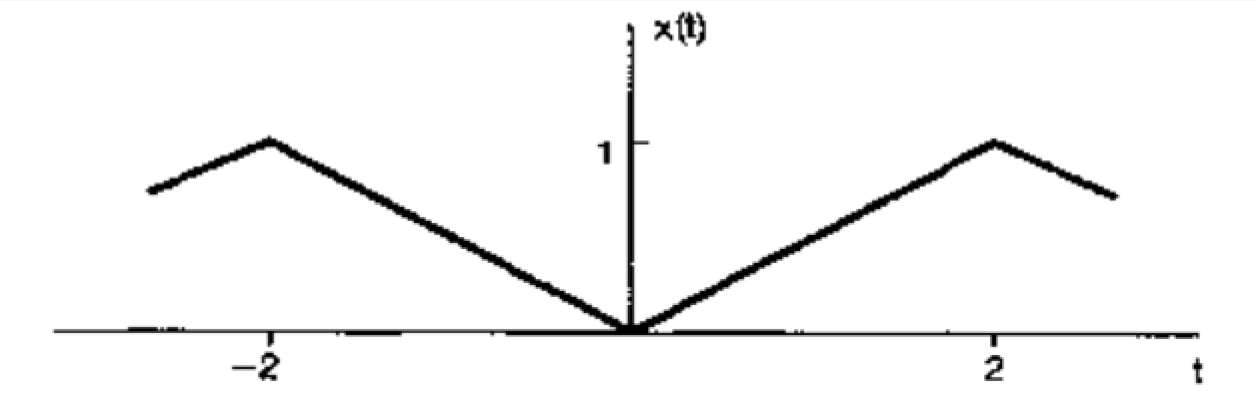
- FS coefficients for $x_2(t) = \frac{1}{2}$ are
- $c_k = \begin{cases} 0 & k \neq 0 \\ \frac{1}{2} & k = 0 \end{cases}$

- Now for $g(t) = x(t-1) - \frac{1}{2}$

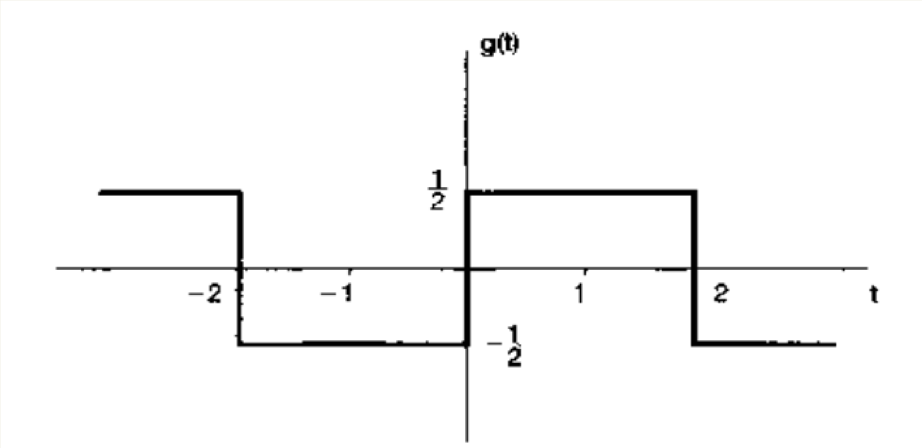
- $d_k = \begin{cases} e^{-jk\frac{\pi}{2}} \frac{1}{k\pi} \sin\left(k\frac{\pi}{2}\right) - 0 & k \neq 0 \\ \frac{1}{2} - \frac{1}{2} & k = 0 \end{cases}$

- $d_k = \begin{cases} e^{-jk\frac{\pi}{2}} \frac{1}{k\pi} \sin\left(k\frac{\pi}{2}\right) & k \neq 0 \\ 0 & k = 0 \end{cases}$

Exp 4 (3.7): Write FS expansion for the signal given below using FS properties



- Differentiate $x(t)$



- $\frac{d}{dt}x(t) = g(t)$

- FS coefficients for $g(t)$,

- $d_k = \begin{cases} e^{-jk\frac{\pi}{2}} \frac{1}{k\pi} \sin\left(k\frac{\pi}{2}\right) & k \neq 0 \\ 0 & k = 0 \end{cases}$

- Now we know
- $\frac{d}{dt}x(t) = g(t) \qquad (i)$
- Using differentiation property
- $z(t) = \frac{d}{dt}x(t) \quad \leftrightarrow \quad Z(k) = jk\omega a_k$
- Eq. (i) in terms of FS coefficients can be written as
- $jk\omega e_k = d_k$
- $e_k = \frac{1}{jk\omega} d_k$
- $e_k = \frac{1}{jk\omega}\left(e^{-jk\frac{\pi}{2}} \frac{1}{k\pi} \sin\left(k\frac{\pi}{2}\right)\right)$

- Now we know
- $e_k = \frac{1}{jk\omega}\left(e^{-jk\frac{\pi}{2}} \frac{1}{k\pi} \sin\left(k\frac{\pi}{2}\right)\right)$
  - $\omega = \frac{2\pi}{T_o}$
  - $T = 2$
  - $\omega = \pi$
- $e_k = \frac{1}{jk\pi}\left(e^{-jk\frac{\pi}{2}} \frac{1}{k\pi} \sin\left(k\frac{\pi}{2}\right)\right)$
- $e_k = \frac{1}{j(k\pi)^2} \sin\left(k\frac{\pi}{2}\right) e^{-jk\frac{\pi}{2}}$

- For $k = 0$,

- Integrate $x(t)$ such that
- $e_o = \frac{1}{T_o} \int x(t)dt$
- e_o =1/2

Thank You !!!